



How We Tested

Our benchmarks comprised SPECviewperf, POV-Ray, Ziff-Davis Multimedia Content Creation and Business Winstone tests, SiSoft Sandra 2004, video encoding tests, multi-tasking tests, Super Pi calculations and gaming tests that included synthetic and real-world games. The benchmarks were in three groups:

The first included SPECviewperf, POV-Ray and Super Pi, which test the Floating-Point Unit (FPU) of the processor. Scores are given in seconds for each of these tests. The second group consisted of the Office Application and Content Creation benchmarks, which included the Ziff-Davis and PC Mark benchmarks.

The last section included gaming, multi-tasking and video encoding benchmarks.

Every benchmark we used is CPU intensive. For example, the POV-Ray and the Super Pi calculations take the same time to complete irrespective of the other components of the system. But results are affected if the memory subsystem, including the hard drive and memory modules, are replaced or changed. With this in mind, we ran the benchmarks for the Intel processors on an Intel D875PBZ with Corsair TWINX DDR 400 512 MB RAM and a Seagate Barracuda 7,200-rpm hard drive. We used the GeForce4 TI 4600 128 MB graphics card, and an Audigy2 sound card. For benchmarking the AMD processors, we used a GigaByte GA-7KNNXP based on the n-Force 2 chipset from nVidia. The video card was again the GeForce TI 4600, and we used the onboard SoundStorm audio chip. The RAM modules and the hard disk remained the same. As expected, the Intel-AMD competition was tough, with pros and cons on both sides.

The benchmarks we used

SPECviewperf: Measures OpenGL 3D rendering. When 3D rendering is involved, you can be sure that there are lots of Floating Point Unit (FPU)-intensive operations, which place a major load on the CPU.

POV-Ray: An open-source ray-tracing software. Ray-tracing involves intensive use of the FPU of the CPU. So, the better the FPU, the faster is the rendering of an image.

Ziff-Davis PC Magazine Content Creation Benchmark: Tests the entire system, with emphasis on the CPU, and uses real-world applications such as Adobe Photoshop and NewTek LightWave.

Ziff-Davis PC Magazine Business Winstone Benchmark: Uses Office applications such as Word and Excel to test performance.

FutureMark PC Mark 2004: This benchmark consists of a separate battery of tests just for the CPU. It uses small chunks of real-world applications such as video encoding and file compression.

SiSoft Sandra 2004: A synthetic benchmark, comprising two tests: the first is the CPU Integer Benchmark that gives us the Dhrystone and Whetstone results of the CPU. The other is the CPU multimedia benchmark, which gives a score by generating a visualization of the Mandelbrot fractal.

Video Encoding Test: Involves the conversion of a 50 MB .vob file to DivX using Dr. DivX. We used the DivX PRO 5.05 codec.

Multi-tasking Test: Involves the video encoding test while opening a 502 MB TIFF file in Adobe Photoshop.

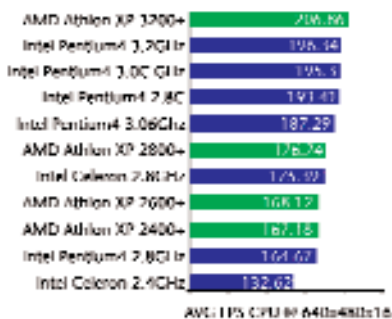
Super Pi Calculation Test: Calculates the value of pi (22/7) to a specifiable number of million floating points. This could be anywhere from two to 32 million. The score is the time taken for calculation.

Aquamark3: The latest gaming benchmark, based on a real-world game engine that puts to test the entire graphics and memory subsystem. After running it, CPU-specific FPS scores can be obtained. These are taken as the scores.

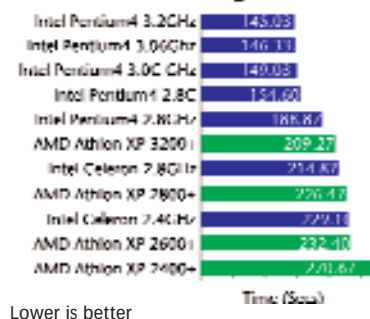
UT 2003: A real-world game benchmark based on the DX 8.1 API that runs different game scenes to give a combined score in frames per second (FPS).

Serious Sam 2: A real-world OpenGL game benchmark which results in an FPS score.

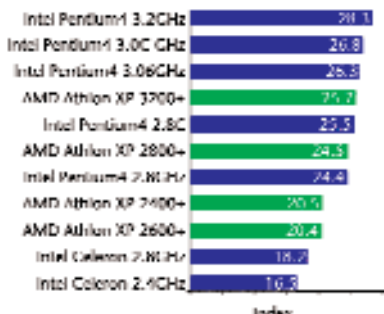
AquaMark 3 Scores



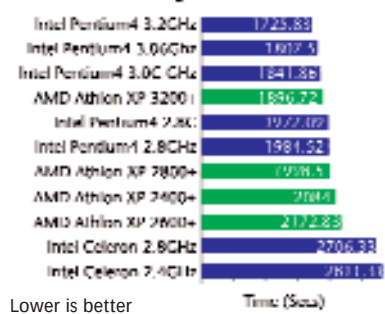
Multi-tasking Scores



Content Creation 2004



PovRay 3.5 Scores



Conclusion

In the end, it boils down to how the CPUs fared against each other. The differences in architecture are apparent, since each manufacturer implements clock speeds and FSB speeds differently. The L1 and L2 caches, too, come into play. Both manufacturers have made cache as well as FSB changes on their processors, for performance gains.

A Celeron running at 400 MHz FSB with 128 KB of L2 cache will not perform as well as a Pentium 4, because of the lower FSB and the smaller caches, even if the clock speed is the same. And that's where the price factor rears its head.

The decision is rather simple, really. The Celerons are a viable option for a Rs 20,000 computer, one that is not often used for heavy data processing applications say, games or audio and video editing. It's ideal if you only need to work with Office applications and the like—optimisations in the Northwood core won't give you a performance increase in